

Decomposition of Simple All Small Games

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Abstract

Simple all small games are games in which both players have exactly one option from every local non-terminal position. Each simple all small game born on day n corresponds to a binary tree with depth n where each non-terminal node has exactly two child nodes. We focus our study here on simple all small games with integer valued atomic weights. An all small game G is called a molecule if G has integer valued atomic weight. We found out there are 25 distinct molecules born on or before day 5, moreover, all the molecules can be decomposed as a sum of 13 elementary games with atomic weight 1 or 0.

1 Introduction

We assume our reader is familiar with the basics of combinatorial game theory as presented in *Winning Ways*[1] or *On Numbers and Games*[2]. Specifically, our reader should be familiar with the following: the abstract definitions of the group of games, the partial order relation of games, the canonical form of games, all small games like the star(*), up($\uparrow$), and down($\downarrow$), and the atomic weight calculus.

We will focus our discussion here on all small games in which both players have exactly one option from every local non-terminal position. These games are named *simple all small games*. Since both players have exactly one option from every local non-terminal position, each simple all small game corresponds to a binary tree within which each node is either a terminal node or a node with exactly one left and one right branch. For example, $*$ = {0|0} and $\uparrow$ = {0|*} are simple all small games because each player has exactly one option from every local non-terminal position. As a counter example, $*2$ = {0, * | 0, *} is not a simple all small game, because the players have more than one option from $*2$.

An all small game G is called a *molecule* if G has integer valued atomic weight, and an *atom* if G has atomic weight one. Many molecules can be

decomposed as sums of atoms and *. One famous example is the up-star equality: $\{0|\uparrow\} = \uparrow + \uparrow + *$, note that $\{0|\uparrow\}$ is a molecule with atomic weight 2, the right-hand side is a sum of atoms $\uparrow$ and *. In this article, we shall examine all the simple molecules born on or before day 5, and check whether these molecules can be decomposed as a sum of atoms and star.

The remaining part of this section introduces the notations and conventions used in this article.

Let G be a game. We use the notation $[G]$ to denote the atomic weight of G .

$$[G] \Leftrightarrow \text{the atomic weight of } G \dots \dots \dots (1)$$

$\nabla^L(G)$ and $\nabla^R(G)$ denote the Left and the Right incentive of G .

$$\nabla^L(G) = G^L - G \dots \dots \dots (2)$$

$$\nabla^R(G) = G - G^R \dots \dots \dots (3)$$

$n.G$, where n is an integer, denotes the sum of n copies of G .

$$n.G = \underbrace{G + \dots + G}_{n \text{ times}} \dots \dots \dots (4)$$

Let G and H be two games. We write $G \ll H$ if the sum of any copies of G is less than H .

$$G \ll H \Leftrightarrow n.G < H, \text{ for any integer } n \dots (5)$$

2 Star Atoms and Molecule Calculus

In this section, we introduce some common atoms before proceeding to examine the molecules born on or before day 5.

Definition 1: (*ups and downs*)

$$\uparrow_0 = 0 \dots \dots \dots (7)$$

$$\uparrow_n = \{\uparrow_{n-1} | *\}, n > 0 \dots \dots \dots (8)$$

Each $\uparrow_n$ is called an *up*.

We use the notation $\uparrow$ to stand for $\uparrow_1$. The negation of up is called *down*.

$$\downarrow_n = -\uparrow_n, n \geq 0. \dots\dots\dots(9)$$

The family of ups has the following properties:

Proposition 1: for $n > 0$,

$$[\uparrow_n] = 1 \dots\dots\dots(9)$$

$$\uparrow_n + \uparrow_{n-1} \uparrow_{n-1} > * \dots\dots\dots(10)$$

$$\uparrow_{n+1} \uparrow_n < \uparrow_n \uparrow_{n-1} \dots\dots\dots(11)$$

Note that (10) implies

$$\nabla^R(\uparrow_n) > \nabla^L(\uparrow_n) \dots\dots\dots(12)$$

And, (11) implies ups are well ordered.

Definition 2: (*twin-ups and twin-downs*)

$$\uparrow_{ab} = \{\uparrow_a + \uparrow_b \mid *\}, a \geq b > 0. \dots\dots\dots(13)$$

Each $\uparrow_{ab}$ is called a *twin-up*. We use the notation $\uparrow$ to stand for $\uparrow_{11}$. The negation of twin-up is called *twin-down*, denoted as $\downarrow_{ab}$.

$$\downarrow_{ab} = -\uparrow_{ab} \dots\dots\dots(14)$$

Twin ups have the following properties:

Proposition 2: for $a \geq b > 0, c \geq d > 0$

$$[\uparrow_{ab}] = 1 \dots\dots\dots(15)$$

$$\uparrow_{ab} + \uparrow_{ab} \uparrow_a \uparrow_b > * \dots\dots\dots(16)$$

$$\uparrow_{ab} > \uparrow_{cd}, \text{ when } b > d \dots\dots\dots(17)$$

$$\uparrow_{ab} > \uparrow_{cb}, \text{ when } b=d, a > c \dots\dots\dots(18)$$

Note that (16) implies

$$\nabla^R(\uparrow_{ab}) > \nabla^L(\uparrow_{ab}) \dots\dots\dots(19)$$

(17) and (18) implies twin-ups are well ordered.

Definition 3: (*star atom*)

An atom A is called a *star atom* if $A^R = *$, and $\nabla^R(A) > \nabla^L(A)$.

Clearly, ups and twin-ups are star atoms.

Definition 4: (*monotonic molecule and maximum atom*)

A molecule M is *monotonic* if M can be written as:

$$M = \sum_{i=1}^n c_i \cdot A_i + c_0 \cdot *, \dots\dots\dots(20)$$

where $A_1 < A_2 < \dots < A_n$ are star atoms, $c_n > 0$, $c_0 = 0$ or 1 . A_n is called the *maximum atom* of M .

Note that if M is monotonic then $-M$ is not monotonic, because we require $c_n > 0$.

Since $\nabla^R(A_n) > \nabla^R(A_k) > \nabla^R(-A_k)$, for $k < n$, we have $\nabla^R(M) = \nabla^R(A_n)$. Also since $\nabla^R(A_n) > \nabla^L(A_n)$ and $\nabla^R(A_n) > \nabla^L(-A_k) > \nabla^L(A_k)$, for $k < n$, We have $\nabla^R(A_n) > \nabla^L(M)$. The next proposition summarizes the result.

Proposition 3: (*maximum atom incentive*)

If M be a monotonic molecule with maximum atom A , then

$$\nabla^R(M) = \nabla^R(A) > \nabla^L(M). \dots\dots\dots(21)$$

According to proposition 3, when playing a sum of ordered star atoms, one should remove the opponent's largest atom unless he has a larger atom. The next theorem is an important tool to analyze molecules.

Theorem: (*molecule calculus*)

If $G^L \geq 0$, $G^R \triangleright 0$ is a monotonic molecule with maximum atom m , and

$$G \geq G^R + m + *,$$

then

$$G = G^R + x + *, \dots\dots\dots(22)$$

where

$$x = \{m^L, G^L - G^R + * \mid *\} \dots\dots\dots(23)$$

Proof: We want to show that the first player cannot win the sum $S = G^R + x + * - G$.

- L cannot win S by a move from $-G$ to $-G^R$, because R can follow a move from x to $*$.
- L cannot win S by a move from x to $G^L - G^R + *$, because R can follow a move from $-G$ to $-G^L$.
- L cannot win S by a move from G^R . By proposition 4, $\nabla^L(G^R) < \nabla^R(m) = m + * \leq G - G^R = \nabla^R(G) = \nabla^L(-G)$, so a move from G^R will not be better than a move from $-G$.
- L cannot win S by a move from x to m^L , because $m^L - x \leq m^L - m = \nabla^L(m)$, so the move will not be better than a move from G^R to $G^R - m + m^L$.
- R cannot win S by a move from x to $*$, because L can follow a move from $-G$ to $-G^R$.
- R cannot win S by a move from $-G$ to $-G^L$, because L can follow a move from x to $G^L - G^R + *$.
- R cannot win S by a move from G^R , because $\nabla^R(G^R) = \nabla^R(m) = m + * \leq x + * = \nabla^R(x)$, ($x \geq m$), so the move will not be better than a

move from x to $*$.

Thus, $S = 0$ and $G = G^R + x + *$.

Moreover, if $[G^L - G^R] \leq 2$ then $[x] = 1$, thus G is a monotonic molecule with maximum star atom x .
□

3 Molecules Born on Day 0, 1, 2 and 3

This section enumerates all the simple all small games born on day 0, 1, 2 and 3. Note that for simple all small games other than 0, there is exactly one game in both the left and the right sets.

On day 0, we have the simplest game

$$0 = \{\} \dots \dots \dots (G-0)$$

where both the left and right are empty.

On day 1, there is only one new simple game:

$$* = \{0|0\} \dots \dots \dots (G-1)$$

$*$ has atomic weight 0, and the property $* + * = 0$.

On day 2, there is only one new simple game:

$$\uparrow = \{0|*\} \dots \dots \dots (G-2)$$

$\uparrow$ has atomic weight 1 and is the simplest atom. We ignore the negations and only consider games not less than 0.

On day 3, there are 4 possible new simple games:

Table 1. simple all small games born on day 3.

$G^L \backslash G^R$	0	*	$\uparrow$
0	~	~	$\{0 \uparrow\}$
$\uparrow$	$\{\uparrow 0\}$	$\{\uparrow *\}$	$\{\uparrow \uparrow\}$

~ denotes games born before day 3.

By replacing with reversible move, one can get:

$$\{\uparrow|0\} = *,$$

By applying the molecule calculus to the column $G^R = \uparrow$, one can get

$$\{0|\uparrow\} = \uparrow\uparrow*.$$

$$\{\uparrow|\uparrow\} = \uparrow\uparrow*.$$

So there are only 2 new games on day 3:

$$\uparrow_2 = \{\uparrow|*\} \dots \dots \dots (G-3)$$

$$\uparrow\uparrow* \dots \dots \dots (G-4)$$

Note that (G-0) to (G-4) are molecules.

4 Molecules Born on Day 4

This section enumerates all the simple all small games born on day 4. Up to day 3, there are 5 simple all small games. These 5 simple all small games will be served as the right options of the possible molecules born on or day 4. 4 of the 5 games are greater than or equal to 0. These 4 games will be served as the left option of the possible new simple all small games born on day 4.

On day 4, there are 14 possible simple all small games (excluding games born up to day 3).

Table 2. simple all small games born on day 4.

$G^L \backslash G^R$	0	*	$\uparrow$
0	~	~	~
$\uparrow$	~	~	~
$\uparrow_2$	$\{\uparrow_2 0\}$	$\{\uparrow_2 *\}$	$\{\uparrow_2 \uparrow\}$
$\uparrow\uparrow*$	$\{\uparrow\uparrow* 0\}$	$\{\uparrow\uparrow* *\}$	$\{\uparrow\uparrow* \uparrow\}$

$G^L \backslash G^R$	$\uparrow_2$	$\uparrow\uparrow*$
0	$\{0 \uparrow_2\}$	$\{0 \uparrow\uparrow*\}$
$\uparrow$	$\{\uparrow \uparrow_2\}$	$\{\uparrow \uparrow\uparrow*\}$
$\uparrow_2$	$\{\uparrow_2 \uparrow_2\}$	$\{\uparrow_2 \uparrow\uparrow*\}$
$\uparrow\uparrow*$	$\{\uparrow\uparrow* \uparrow_2\}$	$\{\uparrow\uparrow* \uparrow\uparrow*\}$

~ denotes games born before day 4.

By replacing with reversible move, one can get:

$$\{\uparrow_2|0\} = *,$$

$$\{\uparrow\uparrow*|*\} = \uparrow.$$

The molecule calculus can be applied to the columns $G^R = \uparrow, \uparrow_2$, and $\uparrow\uparrow*$. We have

$$\{\uparrow_2|\uparrow\} = \uparrow\uparrow*.$$

$$\{\uparrow\uparrow*|\uparrow\} = \uparrow_2\uparrow*.$$

$$\{0|\uparrow_2\} = \uparrow_2\uparrow_2*.$$

$$\{\uparrow|\uparrow_2\} = \uparrow_2\uparrow_2*.$$

$$\{\uparrow_2|\uparrow_2\} = \uparrow_2\uparrow_2*.$$

$$\{\uparrow\uparrow*|\uparrow_2\} = \uparrow_2\uparrow_2*.$$

$$\{0|\uparrow\uparrow*\} = \uparrow\uparrow\uparrow$$

$$\{\uparrow|\uparrow\uparrow*\} = \uparrow\uparrow\uparrow$$

$$\{\uparrow_2|\uparrow\uparrow*\} = \uparrow\uparrow\uparrow$$

$$\{\uparrow\uparrow*|\uparrow\uparrow*\} = \uparrow\uparrow\uparrow$$

So there are only 5 new molecules born on day 4:

$$\uparrow_3 = \{\uparrow_2 | *\} \dots\dots\dots (G-5)$$

$$\uparrow\uparrow * = \{\uparrow\uparrow * | 0\} \dots\dots\dots (G-6)$$

$$\uparrow_2\uparrow^* \dots\dots\dots (G-7)$$

$$\uparrow_2\uparrow_2^* \dots\dots\dots (G-8)$$

$$\uparrow\uparrow\uparrow \dots\dots\dots (G-9)$$

5 Molecules Born on Day 5

This section enumerates all the simple all small games born on day 5. Up to day 4, there are 10 simple molecules. The right options of the possible all small games born on day 5 including, in addition to the 10 molecules, the molecule $\Downarrow^*$. 8 of the 10 molecules born up to day 4 are greater than or equal to 0. These 8 molecules will be served as the left option of the possible molecules born on day 5.

On day 5, there are 56 possible simple all small games (excluding games born up to day 4).

Table 3. simple all small games born on day 5.

$G^L \backslash G^R$	0	*	$\uparrow$
0	$\sim$	$\sim$	$\sim$
$\uparrow$	$\sim$	$\sim$	$\sim$
$\uparrow_2$	$\sim$	$\sim$	$\sim$
$\uparrow_3$	$\{\uparrow_3 0\}$	$\{\uparrow_3 *\}$	$\{\uparrow_3 \uparrow\}$
$\uparrow\uparrow^*$	$\sim$	$\sim$	$\sim$
$\uparrow_2\uparrow^*$	$\{\uparrow_2\uparrow^* 0\}$	$\{\uparrow_2\uparrow^* *\}$	$\{\uparrow_2\uparrow^* \uparrow\}$
$\uparrow_2\uparrow_2^*$	$\{\uparrow_2\uparrow_2^* 0\}$	$\{\uparrow_2\uparrow_2^* *\}$	$\{\uparrow_2\uparrow_2^* \uparrow\}$
$\uparrow\uparrow\uparrow$	$\odot$	$\odot$	$\{\uparrow\uparrow\uparrow \uparrow\}$

$G^L \backslash G^R$	$\uparrow_2$	$\uparrow_3$
0	$\sim$	$\{0 \uparrow_3\}$
$\uparrow$	$\sim$	$\{\uparrow \uparrow_3\}$
$\uparrow_2$	$\sim$	$\{\uparrow_2 \uparrow_3\}$
$\uparrow_3$	$\{\uparrow_3 \uparrow_2\}$	$\{\uparrow_3 \uparrow_3\}$
$\uparrow\uparrow^*$	$\sim$	$\{\uparrow\uparrow^* \uparrow_3\}$
$\uparrow_2\uparrow^*$	$\{\uparrow_2\uparrow^* \uparrow_2\}$	$\{\uparrow_2\uparrow^* \uparrow_3\}$
$\uparrow_2\uparrow_2^*$	$\{\uparrow_2\uparrow_2^* \uparrow_2\}$	$\{\uparrow_2\uparrow_2^* \uparrow_3\}$
$\uparrow\uparrow\uparrow$	$\{\uparrow\uparrow\uparrow \uparrow_2\}$	$\{\uparrow\uparrow\uparrow \uparrow_3\}$

$G^L \backslash G^R$	$\uparrow\uparrow^*$	$\uparrow\uparrow\uparrow$
0	$\{0 \uparrow\uparrow^*\}$	$\{0 \uparrow\uparrow\uparrow\}$
$\uparrow$	$\{\uparrow \uparrow\uparrow^*\}$	$\{\uparrow \uparrow\uparrow\uparrow\}$
$\uparrow_2$	$\{\uparrow_2 \uparrow\uparrow^*\}$	$\{\uparrow_2 \uparrow\uparrow\uparrow\}$
$\uparrow_3$	$\{\uparrow_3 \uparrow\uparrow^*\}$	$\{\uparrow_3 \uparrow\uparrow\uparrow\}$
$\uparrow\uparrow^*$	$\{\uparrow\uparrow^* \uparrow\uparrow^*\}$	$\{\uparrow\uparrow^* \uparrow\uparrow\uparrow\}$
$\uparrow_2\uparrow^*$	$\{\uparrow_2\uparrow^* \uparrow\uparrow^*\}$	$\{\uparrow_2\uparrow^* \uparrow\uparrow\uparrow\}$
$\uparrow_2\uparrow_2^*$	$\{\uparrow_2\uparrow_2^* \uparrow\uparrow^*\}$	$\{\uparrow_2\uparrow_2^* \uparrow\uparrow\uparrow\}$
$\uparrow\uparrow\uparrow$	$\{\uparrow\uparrow\uparrow \uparrow\uparrow^*\}$	$\{\uparrow\uparrow\uparrow \uparrow\uparrow\uparrow\}$

$G^L \backslash G^R$	$\uparrow_2\uparrow^*$	$\uparrow_2\uparrow_2^*$
0	$\{0 \uparrow_2\uparrow^*\}$	$\{0 \uparrow_2\uparrow_2^*\}$
$\uparrow$	$\{\uparrow \uparrow_2\uparrow^*\}$	$\{\uparrow \uparrow_2\uparrow_2^*\}$
$\uparrow_2$	$\{\uparrow_2 \uparrow_2\uparrow^*\}$	$\{\uparrow_2 \uparrow_2\uparrow_2^*\}$
$\uparrow_3$	$\{\uparrow_3 \uparrow_2\uparrow^*\}$	$\{\uparrow_3 \uparrow_2\uparrow_2^*\}$
$\uparrow\uparrow^*$	$\{\uparrow\uparrow^* \uparrow_2\uparrow^*\}$	$\{\uparrow\uparrow^* \uparrow_2\uparrow_2^*\}$
$\uparrow_2\uparrow^*$	$\{\uparrow_2\uparrow^* \uparrow_2\uparrow^*\}$	$\{\uparrow_2\uparrow^* \uparrow_2\uparrow_2^*\}$
$\uparrow_2\uparrow_2^*$	$\{\uparrow_2\uparrow_2^* \uparrow_2\uparrow^*\}$	$\{\uparrow_2\uparrow_2^* \uparrow_2\uparrow_2^*\}$
$\uparrow\uparrow\uparrow$	$\{\uparrow\uparrow\uparrow \uparrow_2\uparrow^*\}$	$\{\uparrow\uparrow\uparrow \uparrow_2\uparrow_2^*\}$

$G^L \backslash G^R$	$\uparrow\uparrow\uparrow$	$\Downarrow^*$
0	$\{0 \uparrow\uparrow\uparrow\}$	$\{0 \Downarrow^*\}$
$\uparrow$	$\{\uparrow \uparrow\uparrow\uparrow\}$	$\{\uparrow \Downarrow^*\}$
$\uparrow_2$	$\{\uparrow_2 \uparrow\uparrow\uparrow\}$	$\{\uparrow_2 \Downarrow^*\}$
$\uparrow_3$	$\{\uparrow_3 \uparrow\uparrow\uparrow\}$	$\{\uparrow_3 \Downarrow^*\}$
$\uparrow\uparrow^*$	$\{\uparrow\uparrow^* \uparrow\uparrow\uparrow\}$	$\odot$
$\uparrow_2\uparrow^*$	$\{\uparrow_2\uparrow^* \uparrow\uparrow\uparrow\}$	$\odot$
$\uparrow_2\uparrow_2^*$	$\{\uparrow_2\uparrow_2^* \uparrow\uparrow\uparrow\}$	$\odot$
$\uparrow\uparrow\uparrow$	$\{\uparrow\uparrow\uparrow \uparrow\uparrow\uparrow\}$	$\odot$

$\sim$ denotes games born before day 5.

$\odot$ denotes games that are switches of molecules.

The molecule calculus can be applied to the columns:

$$G^R = \uparrow, \uparrow_2, \uparrow_3, \uparrow\uparrow^*, \uparrow\uparrow\uparrow, \uparrow_2\uparrow^*, \uparrow_2\uparrow_2^* \text{ and } \uparrow\uparrow\uparrow.$$

In the above columns, there are 9 new molecules:

$$\uparrow_3\uparrow^* \dots\dots\dots (G-10)$$

$$\uparrow_3\uparrow_2^* \dots\dots\dots (G-11)$$

$$\uparrow_3\uparrow_3^* \dots\dots\dots (G-12)$$

$$\{\uparrow_2\uparrow_2 | *\} + \uparrow^* \dots\dots\dots (G-13)$$

$$\uparrow_2\uparrow\uparrow \dots\dots\dots (G-14)$$

$$\uparrow_2\uparrow_2\uparrow \dots\dots\dots (G-15)$$

$$\uparrow_2\uparrow_2\uparrow_2 \dots\dots\dots (G-16)$$

$$\uparrow\uparrow\uparrow \dots\dots\dots (G-17)$$

$$\uparrow\uparrow\uparrow^* \dots\dots\dots (G-18)$$

In the other columns, there are 7 new molecules.

$$\{0 | \Downarrow^*\} \dots\dots\dots (G-19)$$

$$\{\uparrow\uparrow\uparrow^* | \Downarrow^*\} + \Downarrow^* = \{\uparrow | \Downarrow^*\} \dots\dots\dots (G-20)$$

$$\{\uparrow\uparrow\uparrow_2^* | \Downarrow^*\} + \Downarrow^* = \{\uparrow_2 | \Downarrow^*\} \dots\dots\dots (G-21)$$

$$\{\uparrow\uparrow\uparrow_3^* | \Downarrow^*\} + \Downarrow^* = \{\uparrow_3 | \Downarrow^*\} \dots\dots\dots (G-22)$$

$$\uparrow_4 = \{\uparrow_3 | *\} \dots\dots\dots (G-23)$$

$$\uparrow_{22}^* = \{\uparrow_2\uparrow_2^* | 0\} \dots\dots\dots (G-24)$$

$$\uparrow_{21}^* = \{\uparrow_2\uparrow^* | 0\} \dots\dots\dots (G-25)$$

Thus, the total number of none-zero molecules born on or before day 5 is 25.

In addition to the 25 molecules, there are 6 none-molecule games (marked $\odot$) in table 3. All these none-molecule games are molecule switches.

$\{\uparrow\uparrow\uparrow|0\}$(S-1)

$\{\uparrow\uparrow\uparrow|\ast\}$(S-2)

$\{\uparrow\uparrow\ast|\downarrow\ast\}$(S-3)

$\{\uparrow_2\uparrow\ast|\downarrow\ast\}$(S-4)

$\{\uparrow_2\uparrow_2\ast|\downarrow\ast\}$(S-5)

$\{\uparrow\uparrow\uparrow|\downarrow\ast\}$(S-6)

Since the incentive of any molecule switch is greater than the incentive of any molecule, when playing sums of molecules and molecule switches, molecule switches should always be played before molecules.

6 Atoms Born on or before Day 5

A close look at the molecules G-1 to G-25 will found out that these molecules are a sum of 13 elementary games:

$\{0|\downarrow\ast\}$ (E-0)

$\uparrow$(E-1)

$\uparrow_2$(E-2)

$\uparrow_3$(E-3)

$\uparrow_4$(E-4)

$\{\uparrow_2\uparrow_2\downarrow|\ast\}$(E-5)

$\uparrow\uparrow$(E-6)

$\uparrow\uparrow_{21}$(E-7)

$\uparrow\uparrow_{22}$(E-8)

$\{\uparrow\uparrow+\uparrow|\ast\}$(E-9)

$\{\uparrow\uparrow+\uparrow_2|\ast\}$(E-10)

$\{\uparrow\uparrow+\uparrow_3|\ast\}$(E-11)

$\ast$(E-12)

The first 12 elementary games are totally ordered:

$$(E-0) < (E-1) < (E-2) < \dots < (E-11). \dots(24)$$

The last elementary game $\ast$ (E-12) is confused with all other elementary game.

Note that (E-1) to (E-11) are star atoms, so any sum of (E-1) to (E-12) is a monotonic molecule and thus proposition 3 can be applied. According to proposition 3, when playing a sum of ordered star atoms, one should remove the opponent's largest atom unless he has a larger atom.

(E-0) has atomic weight 0 and is called *tiny*.

The right incentives of these elementary games are totally ordered:

$$\begin{aligned} & \nabla^R(E-12) \\ & < \nabla^R(E-1) \\ & < \nabla^R(E-2) \\ & < \dots \\ & < \nabla^R(E-6) \\ & < \nabla^R(E-0) \\ & < \nabla^R(E-7) \\ & < \nabla^R(E-8) \\ & < \dots \\ & < \nabla^R(E-11). \dots(25) \end{aligned}$$

When playing a sum of these elementary games, one should move at the opponent's elementary game with the largest right incentive unless he has an elementary game with larger right incentive. In the later case, if there is an odd number of $\ast$, then play at $\ast$. Otherwise, there must be an even number of $\ast$, he should play his elementary game with the largest right incentive.

By repeating the above playing strategy, we can determine the outcome of any sum of E-0 to E-12.

7 Conclusion

Atoms are the building elements of complex games. In this paper, we have shown that there are only 11 distinct atoms born on or before day 5. We introduced the molecule calculus to help the decomposition of molecules. All the molecules with positive atomic weights born on or before day 5 can be decomposed as a sum of these atoms and $\ast$. These atoms are totally ordered, and thus the optimal strategy for playing sums of these atoms can be easily derived.

References

- [1] E. R. Berlekamp, J. H. Conway, R. K. Guy, "Winning Ways", Academic Press, New York, 1982
- [2] J. H. Conway, "On Numbers and Games", Academic Press, New York, 1976
- [3] K. Y. Kao, "Sumers – Sums of Ups and Downs ", Integers, Electronic Journal of Combinatorial Number Theory, Vol.5 2005